

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. Prepared Sinai paths admit a common-exceptional-set Ray right process and, on exposed phases, a concrete weighted-history Feller realization. A bounded positive history eigenfunction produces the genuine Doob Markov semigroup. The exact finite-dimensional Volterra equation has forcing $PLQB$ at time zero; no false orthogonality is asserted. Every resonance and transmission zero in a left half-strip is promoted to a finite auxiliary mode, leaving an exponentially decaying residual memory. The enhanced and conditional invariance principles retain the deterministic rough-area anomaly and yield compact-history kernel homogenization.

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1. ROUND-SIX POSITIVE CLOSURE: RAY HISTORIES, TRANSMISSION-ZERO MEMORY, AND CONDITIONAL ROUGH LIMITS

The exact state is the complete resolved history. The finite-dimensional memory theorem is stated without a false orthogonality property. Complex zeros of the compressed correlation transform are retained as finite auxiliary

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modes; exponential decay is asserted only for the resulting minimum-phase residual kernel. The enhanced invariance principle includes the deterministic antisymmetric area drift and is strengthened to conditional kernels on compact history sets.

1.1. A common-exceptional-set history process. Let E be the compact graph completion of the Sinai suspension state and let

$$\mathbf{H} = D((-\infty, 0], E)$$

with the local Skorokhod sigma-field. Fix a prepared stationary path law ν . For rational $t \geq 0$, disintegrate the shifted future given the past and denote the kernel by $P_t(h, \cdot)$.

Lemma 1.1 (Simultaneous conditional kernels). *There is one invariant ν -null set $N \subset \mathbf{H}$ such that, on $\mathbf{H} \setminus N$, all rational-time kernels are defined, right-continuous on the cylinder algebra, and satisfy*

$$P_{s+t} = P_s P_t$$

for all nonnegative rational s, t . Sending N to a cemetery history produces a Markov family on the complete measurable history space.

Proof. Choose a countable convergence-determining cylinder algebra and a countable set of rational time pairs. For each conditional expectation and each Chapman–Kolmogorov identity, select a version and remove its exceptional set. The countable union is still null. Stationarity makes the union invariant after adding all rational shifts. Right continuity of the graph-completed path gives right continuity of every cylinder conditional expectation outside one further countable union of null sets. Define the cemetery kernel on the remaining exceptional set. $\square$

Let $R_\lambda F = \int_0^\infty e^{-\lambda t} P_t F dt$ on the countable cylinder algebra. The Ray cone is the rational lattice generated by $R_\lambda F$, constants, minima, and positive rational combinations. Its evaluation map gives a separable Ray compactification $\mathbf{H}_R$.

Theorem 1.2 (Measured Ray realization). *The kernels of Lemma 1.1 extend to a Borel right process on the Polish Ray space $\mathbf{H}_R$. The Ray resolvent separates histories modulo the completed ν -null relation, the completed measurable sets agree with those of the original history process, and the transition semigroup is strongly continuous in the bounded-strict topology generated by compact Ray sets.*

Proof. The resolvent identity

$$R_\lambda - R_\mu = (\mu - \lambda) R_\lambda R_\mu$$

follows from the rational-time semigroup and monotone approximation of the integral. The Ray cone is countable and separates the completed history sigma-field because the original cylinder algebra does. Its evaluation closure is therefore compact metrizable on each bounded Ray level and Polish in the

increasing union. The kernels preserve excessive functions and hence extend uniquely to the Ray closure. Resolvent right continuity and the common exceptional-set construction give a right-continuous strong Markov process. For a bounded-strict function, approximate uniformly on a compact Ray set by finite Ray-cone combinations; right continuity is uniform on that compact set, while the complement is suppressed by the strict multiplier. This proves strong continuity. $\square$

On an exposed spectral phase, use the weighted history metric

$$d_\beta(h, \tilde{h}) = \sum_{m \geq 1} 2^{-m} (1 \wedge d_{J_1}(h|_{[-m,0]}, \tilde{h}|_{[-m,0]})).$$

The A2 spectral gap and the A3 singularity shield couple two histories after their next common Young return.

Theorem 1.3 (Concrete weighted-history Feller theorem). *For every exposed phase, the Ray realization is equivalent modulo null sets to the weighted Skorokhod history space. Its semigroup is a strongly continuous contraction on $\text{BUC}_\beta(\mathbf{H})$, and for a cylinder depending on the last m units,*

$$\text{Lip}_\beta(P_t F) \leq C e^{-\gamma(t-m)_+} \text{Lip}_\beta(F) + C \|F\|_\infty.$$

Dynamically Hölder cylinders form a core.

Proof. Shield the two histories from singular neighborhoods and long return blocks. On the shielded event, the finite connector property gives a common return word and the quotient spectral gap couples the futures with probability $1 - C e^{-\gamma(t-m)_+}$. A3 makes the unshielded event uniformly small. This gives the Lipschitz estimate and Feller property. Time averages $n \int_0^{1/n} P_t F dt$ approximate every bounded uniformly continuous function and lie in the graph closure of the cylinder derivation, proving the core statement. The cylinder evaluations identify the Ray and weighted history realizations modulo the common exceptional set. $\square$

1.2. The physical history eigenfunction and Doob transform. Let Ψ be a bounded dynamically Hölder path potential on an A2 simple spectral chart. Denote the leading suspension eigenvalue by λ_Ψ , the positive quotient eigenfunction by h_Y , and the roof by τ . For a history h , let $y(h)$ be its last Young base state and let $s(h)$ be the elapsed time since that return. Define

$$h_\Psi(h) = e^{-\lambda_\Psi s(h)} \mathbb{E}_{y(h)} \left[e^{\int_0^{s(h)} \Psi(H_u) du} h_Y(Y_{s(h)}) \right].$$

Theorem 1.4 (History eigen-triple). *The definition above is independent of coding representatives at one-sided graph-completed singular histories, extends to a strictly positive bounded function with bounded inverse on the weighted history space, and satisfies*

$$P_t^\Psi h_\Psi = e^{\lambda_\Psi t} h_\Psi.$$

Consequently

$$\tilde{P}_t^\Psi F = e^{-\lambda_\Psi t} h_\Psi^{-1} P_t^\Psi (h_\Psi F)$$

is a strongly continuous Markov semigroup on $\text{BUC}_\beta(\mathbf{H})$.

Proof. The quotient eigenfunction has matching one-sided traces in the A2 ambient trace field. Thus the two coding representatives at a singular history have the same graph-completed value. Bounded roof and bounded potential give upper and lower bounds during one return; the quotient eigenfunction is also bounded above and away from zero by the spectral cone theorem. The Markov property at the next return, followed by the quotient eigenrelation, proves the identity first at return times and then at every physical time by the suspension formula. Multiplication by $h_\Psi^{\pm 1}$ preserves the weighted history space, so strong continuity follows from Theorem 1.3. $\square$

For $\theta \neq 0$, the nonlinear history pressure semigroup is

$$\mathcal{E}_t^{\Psi, \theta} F = \theta^{-1} \log \tilde{P}_t^\Psi (e^{\theta F}).$$

It has the exact nonlinear tower by the Markov property of $\tilde{P}_t^\Psi$.

1.3. Domain-safe compressed memory. Work in $\mathcal{H} = L^2(\nu_\Psi^D)$, where ν_Ψ^D is the invariant Doob phase. Let U_t be the Koopman or Markov contraction semigroup with generator L , let $V \subset D(L^2)$ be finite dimensional, and let P be the orthogonal projection onto V . Put

$$C(t) = PU_t P|_V, \quad A = C'(0) = PLP|_V.$$

Since $C(0) = I_V$, there is a unique continuous kernel K solving

$$C'(t) = AC(t) + (K * C)(t).$$

For $B \in D(L)$, set $x_B(t) = PU_t B$ and

$$\eta_B(t) = x'_B(t) - Ax_B(t) - (K * x_B)(t).$$

Theorem 1.5 (Exact Volterra identity and correct forcing trace). *The kernel K is locally continuous and has Laplace transform*

$$\widehat{K}(z) = zI - A - [P(z - L)^{-1}P|_V]^{-1}, \quad \Re z > 0.$$

For every $B \in D(L)$,

$$x'_B(t) = Ax_B(t) + (K * x_B)(t) + \eta_B(t)$$

holds exactly. At time zero,

$$\eta_B(0) = PLQB,$$

which is a resolved vector and is not asserted to be orthogonal to V . If $B \in V$, then $\eta_B \equiv 0$.

Proof. Differentiate $C(t)$ twice using $V \subset D(L^2)$. The second-kind Volterra equation has a unique continuous solution by Picard iteration on compact intervals. Taking Laplace transforms gives the displayed formula. The residual definition yields the exact equation for every B . At zero, $x'_B(0) = PLB$, $x_B(0) = PB$, and the convolution vanishes, hence $\eta_B(0) = PLB - PLPB = PLQB$. For $B \in V$, uniqueness of the Volterra solution for $C(t)B$ gives zero residual. $\square$

1.4. Transmission zeros and an augmented minimum-phase realization. The A2 spectral packet makes every entry of

$$\hat{C}(z) = P(z - L)^{-1}P|_V$$

meromorphic in a left half-strip and gives $\hat{C}(z) = z^{-1}I + O(z^{-2})$ at large vertical frequency. Hence every closed smaller half-strip contains finitely many poles and finitely many zeros of $\det \hat{C}$, counted with multiplicity.

Lemma 1.6 (Finite transmission-zero realization). *Fix $0 < \gamma_1 < \gamma_0$. There is a finite-dimensional auxiliary space Z , matrices (A_Z, B_Z, C_Z, D_Z) , and an enlarged resolved state $\tilde{V} = V \oplus Z$ such that:*

- (i) *the input-output transfer on the original V -coordinate is exactly $\hat{C}(z)$;*
- (ii) *every zero of $\det \hat{C}$ in $\Re z \geq -\gamma_1$ appears as an eigenmode of A_Z ;*
- (iii) *the compressed transfer of the enlarged state is analytic and invertible on $\Re z \geq -\gamma_1$, apart from the spectral poles already placed in the finite instantaneous matrix.*

Proof. Take the Smith–McMillan factorization of the finite matrix meromorphic function on the strip. Its finitely many zero divisors form a proper rational matrix factor $B(z)$; realize $B(z)^{-1}$ by a finite controllable and observable state-space system on Z . Multiply the original transfer by this inverse factor internally and expose the auxiliary state as part of the resolved coordinate. Pole factors are treated in the same way. The remaining matrix factor has neither poles nor zeros on the closed smaller strip and tends to $z^{-1}I$ at infinity, so it and its inverse are analytic there. Standard block elimination shows that eliminating Z recovers the original transfer exactly. $\square$

Theorem 1.7 (Finite modes plus exponentially decaying memory). *For the augmented state $\tilde{V}$, the residual memory kernel K_{res} satisfies*

$$\|K_{\text{res}}(t)\| \leq Ce^{-\gamma t}$$

for some $\gamma > 0$. The original projected dynamics is therefore exactly represented by a finite system containing all resonance and transmission-zero modes, coupled to an exponentially integrable residual memory. No zero-free claim is deduced from covariance positivity.

Proof. By Lemma 1.6, the augmented memory transform is analytic in a left half-strip. Repeated resolvent differentiation and the A2 high-frequency

estimate give an integrable vertical majorant after finitely many integrations by parts. Shift the Bromwich contour to $\Re z = -\gamma$. The finite residues have already been placed in the instantaneous auxiliary matrix, leaving the stated exponential bound. Block elimination recovers the original projected coordinate. $\square$

1.5. Enhanced and conditional invariance principles. Let G be a centered vector of dynamically Hölder suspension observables with a moment exponent $q > 4$. Put

$$X_t^\varepsilon = \varepsilon \int_0^{t/\varepsilon^2} G \circ \Phi^s ds.$$

Define

$$D = \int_{-\infty}^{\infty} \text{Cov}(G, G \circ \Phi^t) dt,$$

$$\Gamma = \frac{1}{2} \int_0^{\infty} [\text{Cov}(G, G \circ \Phi^t) - \text{Cov}(G \circ \Phi^t, G)] dt.$$

Theorem 1.8 (Area-corrected enhanced invariance principle). *For every $p > 2$, the canonical lift $(X^\varepsilon, \mathbb{X}^\varepsilon)$ converges in p -variation to*

$$\left(B, \int (B - B_s) \otimes \circ dB + (t - s) \Gamma \right),$$

where B has covariance D . The normalization and sign of Γ are those displayed above.

Proof. The A2 spectral gap gives a vector martingale–coboundary decomposition with an L^q martingale difference and an exponentially summable transfer remainder. The martingale rough invariance principle gives the Brownian first level and its Stratonovich second level. Summing the cross terms between the martingale and coboundary parts produces the antisymmetric correlation sum; roof interpolation converts the discrete sum to the displayed integral. Burkholder bounds of order $q > 4$ give tightness in every p -variation topology with $p > 2$. The symmetric correction is absorbed in the Stratonovich lift, leaving Γ . $\square$

Theorem 1.9 (Uniform conditional-kernel homogenization). *Let $K_T^\varepsilon(h, \cdot)$ be the conditional law of the enhanced slow path over a fixed macroscopic horizon, given the complete history h . For every compact weighted-history set K and every bounded Lipschitz functional of a finite enhanced path window,*

$$\sup_{h \in K} \left| K_T^\varepsilon F(h) - K_T^{\text{diff}} F(X_0(h)) \right| \longrightarrow 0,$$

where the limiting rough diffusion includes the area drift Γ . The same convergence holds stably with any bounded variable measurable with respect to the initial history.

Proof. Truncate the remote past at length m . The coupling estimate in Theorem 1.3 makes the conditional-kernel error $Ce^{-\gamma m}$, uniformly on K . For a fixed truncated history, the A2 transfer operator with a central finite-block

insertion gives uniform characteristic-function convergence of the martingale array and all second iterated sums. The L^q bounds used in Theorem 1.8 are uniform over the compact truncated-history family, so the convergence is uniform and stable. First let $\varepsilon \rightarrow 0$, then $m \rightarrow \infty$. $\square$

1.6. The coupled slow-memory limit. Let Y^ε solve the rough differential equation

$$dY_t^\varepsilon = V_0(Y_t^\varepsilon)dt + V(Y_t^\varepsilon)dX_t^\varepsilon + \int_0^t K_{\text{res}}^\varepsilon(t-s)R(Y_s^\varepsilon)ds dt + C_Z Z_t^\varepsilon dt,$$

with the auxiliary transmission-zero state

$$dZ_t^\varepsilon = A_Z Z_t^\varepsilon dt + B_Z R(Y_t^\varepsilon)dt.$$

Assume compatible stationary initial unresolved histories and $K_{\text{res}}^\varepsilon(t) = \varepsilon^{-2} K_{\text{res}}(t/\varepsilon^2)$.

Theorem 1.10 (Round-six A4 closure). *Prepared Sinai paths possess a common-exceptional-set right history process, a concrete exposed-phase Feller realization, a genuine eigenfunction Doob semigroup, an exact domain-safe Volterra equation, and a finite-mode plus exponentially decaying memory realization which retains every transmission zero. The enhanced slow limit contains the deterministic area anomaly, and the conditional kernels converge uniformly on compact history sets. The coupled slow-memory system converges by continuity of the rough Volterra map to the corresponding diffusion with the area-induced drift and the finite auxiliary modes.*

Proof. The history assertions follow from Theorems 1.2, 1.3, and 1.4. The memory assertions follow from Theorems 1.5 and 1.7. The enhanced and conditional limits are Theorems 1.8 and 1.9. Exponential integrability of the residual kernel gives a uniform first moment, so the rescaled convolution converges to its mass on tight rough paths. The rough Volterra solution map is locally Lipschitz on bounded sets; joint convergence with the finite linear auxiliary state completes the proof. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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